

Endpoint gauges for integer multiplication

Controlled projector blocks and general binary phase residuals

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With substantial OpenAI GPT-6 Astra and Codex assistance

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Abstract

Complementary auxiliary endpoint gauges and a compatible rational basis at $(a, b, k) = (32, 30, 40)$ give a conditional multiplication bound $T(n) = O(n(\log n)^{1-\kappa})$, where $\kappa = 119723/20000000000 = 5.98615 \times 10^{-6}$. The complex interface admits every nondegenerate binary residual, including alternating ones. Exact finite arithmetic certifies the two recurrence moments and the assembly margins. The conclusion retains the pinned upstream analytic and fixed-tape assumptions; the certificates are neither a formal verification nor an implementation of the complete multiplication machine.

1 Scope and attribution

The computational model is a fixed finite-alphabet multitape Turing machine. Every finite network, basis, prime, and tape count is fixed independently of the input length. The bit payload field is $\mathbb{F}_2$; its address geometry is rational and is subsequently reduced at a suitable odd prime. These two fields have different purposes. The complex payload consists of Gaussian dyadics and uses the retained binary address-label interface.

The retained framework is Douglas Colkitt’s compact-control construction [2], with Bortlesboat’s alignment [3], eumemic’s complex producer and Gaussian resampling [4, 6, 7], dleen’s retained totals and stage sharing [5], and Zhihao Chen’s finite-network contribution [8], pinned at 6725c6a17b17871a35353fd29157f4ed851bc114. The upstream multiplication manuscript is pinned at adc7f1241b42e322a6451854ab7e4b4c146bf78a [1]. The partial-swap, endpoint-gauge, and selected retained-triple development is by icekylinx, with substantial OpenAI GPT-6 Astra assistance; review and integration used OpenAI Codex assistance. Prior contributions retain their attribution and licenses in NOTICE and SOURCES.json. The immediate parent is the partial-swap construction at commit f2ab41aebad47861caf6316282c1793e5513845e. It extends icekylinx’s bulk-recursion PR #10 [9], pinned at 62691e395a0458ce089a1c7b5d89e74291e95e29; its parameterized mixed-width and corrected Gaussian arguments are retained below with the new construction and constants.

2 The scalar construction and endpoint gauges

2.1 Exact lower-triangular conjugation

For a rational projector P , define

$$D_P = \begin{pmatrix} I - P & P \\ P & I - P \end{pmatrix}.$$

Then $D_P^2 = I$ and $\text{rank}(D_P - I) = \text{rank } P$. For nondegenerate $U \subseteq V$, their orthogonal projectors satisfy $P_U P_V = P_V P_U = P_U$, whence

$$D_{P_V} D_{P_U} = D_{P_V - P_U}.$$

Thus either orientation of a nested frame edge costs its absolute dimension change. A scalar gate whose incident roles share one frame commutes with that frame's common address permutation.

Write the ordered lower-lower elimination as $P = L\Pi R$, where L, R are invertible lower triangular and Π is a partial permutation. Put

$$C = RL, \quad Q_X = \Pi\Pi^t, \quad Q_Y = \Pi^t\Pi, \quad K = Q_Y C - C Q_X, \quad G = \begin{pmatrix} L & 0 \\ 0 & R^{-1} \end{pmatrix} \begin{pmatrix} I & 0 \\ K & I \end{pmatrix}.$$

Idempotence gives $\Pi C \Pi = \Pi$. Multiplication, using $\Pi Q_Y = \Pi = Q_X \Pi$, yields the exact identity

$$G^{-1} D_P G = \begin{pmatrix} I - Q_X & \Pi \\ \Pi^t & I - Q_Y \end{pmatrix}.$$

Both G and G^{-1} are lower triangular in the physical order of the two banks. The displayed matrix is exactly the collection of cross-bank swaps specified by Π . A contiguous identity block of t pivots is therefore one width- t interchange, surrounded by the retained $O(V)$ ordered-affine passes. No reversal or gathering of separated pivots is assumed. Choose the address prime only after fixing all rational bases and factorizations, excluding their finitely many bad denominators and nonzero minors.

2.2 Scalar map and dirty-scratch wrapper

For $h \in \{32, 30, 40\}$ write $v_h = \binom{h}{3}$. There is one input X_T for each triple. In common-point context i put $y_{ab} = X_{\{i,a,b\}}$ and compute

$$A_i = \sum_{a < b, a, b \neq i} y_{ab}, \quad D_{i,\{a,b\}} = \sum_{u < v, u, v \notin \{i,a,b\}} y_{uv}.$$

The $q_h = 3v_h + h$ designated outputs retain all totals. The local cancellation-free producer pairs consecutive vertices, sums coarse edges between pairs, and puts internal edges into the corresponding coarse vertex weights. Recurse using the selected base-four paired exclusion schedule; leave-one and leave-two group sums, together with the surviving strips, partition each desired exclusion sum. This proves correctness by induction. The order in context i lists intact global pairs first, then the unlisted vertices other than i . Intern equal supports; pair-star supports shared across contexts are identified by their actual common mask and union mask. Retain total ancestors and prune unused nodes.

Let V copy physical inputs into source-use roles, L be the reversible addition/fanout circuit, and J inject the designated outputs. For target T , inject the three side outputs indexed by $i \in T$ and the three totals A_i , $i \in T$. If a source triple is S , the first coefficient is $[|S \cap T| = 1]$, and the second is $|S \cap T| \bmod 2$. Their sum in $\mathbb{F}_2$ equals $[S = T]$. Hence $JLV = I$. For arbitrary initial auxiliary vector z , the chronological schedule

$$L, J, L^{-1}, V, L, J, L^{-1}, V$$

injects $JLz + JL(z + Vx) = x$ and restores z . This description treats J, V as elementary copy/injection sequences, which are self-inverse over $\mathbb{F}_2$. Three alternating data shears exchange the two data banks; use the inverse schedule for the middle invocation. Scratch is restored for every initial value, not only zero.

2.3 Positive labels preserving carrier dependencies

Use $H_h = I - J/9$ on $\mathbb{Q}^h$ and triple indicator t_T . Then

$$\langle t_S, t_T \rangle = |S \cap T| - 1, \quad U_i = \{x : \sum_j x_j = 3x_i\}, \quad x^t H_h x = \sum_{j \neq i} x_j^2 \quad (x \in U_i).$$

Thus U_i is positive definite, each triple line has norm 2, and intersection-one triple lines are orthogonal. A side output receives $U_i \cap \{x_a + x_b = 0\}$, of dimension $h - 2$; a total receives U_i , of dimension $h - 1$.

For an original cancellation-free node with common set C and union M , its envelope is

$$E(C, M) = \{x : \text{supp } x \subseteq M, x_i = t \ (i \in C), \sum_j x_j = 3t\}.$$

Identify addition nodes having equal envelopes for label propagation, keeping sources separate. Find the initial carrier matching using these envelopes and add its continuation arcs as label-only dependencies. They respect envelope inclusion and strict event order. Backward intersection of all reachable output constraints gives

$$W_g = \{x : \sum_j x_j = 3x_a \ (a \in F_g), x_b + x_c = 0 \ (\{b, c\} \in E(G_g))\}.$$

Here F_g lists reachable common points and G_g their excluded-pair graph. Sources keep their triple lines. We have $E_g \subseteq W_g$ and $W_g \subseteq W_{g'}$ along every augmented arc. Every retained node reaches an output, so W_g lies in a positive U_a . Nondegeneracy, nesting, retained matches, and side-output orthogonality follow.

For addition nodes $|F_g| \in \{1, 2\}$ and G_g avoids F_g . A nonbipartite component forces zero; each bipartite component, including isolated vertices, supplies a signed parameter z_j . If its color classes are A_j, B_j , the remaining relation is

$$(3 - |F_g|)t = \sum_j (|A_j| - |B_j|)z_j.$$

Consequently $\dim W_g$ is the number of bipartite components on the coordinates outside F_g . Signed component relations also give an exact rational containment test. Orthogonal complements and tensor products of these nondegenerate spaces are nondegenerate; ambient H_h is nondegenerate since $h \neq 9$. Reversed invocations use orthogonal complement frames, so reverse dependencies remain nested.

2.4 Carrier matching and its rank histogram

A donor is an addition u ; a receiver is an individual later operand use or designated output use. A donor can preserve either operand as its nonpivot and continue that operand's carrier to a later use w when $W_u \subseteq W_w$. A matching joins carrier paths without cycles or simultaneous identification of occupied roles. Every value retains an unmatched first use, which supplies its carrier path. Thus

$$R_h = c_h + q_h - |\mathcal{M}_h|.$$

Use events ordered by new dimension, old envelope dimension, and original node/output number. Deterministic Hopcroft–Karp orders donors by node number, first operand before second, and uses by consumer then output order, in both breadth-first and shortest augmenting-path searches. Initial matching arcs were propagated before enlargement; they remain valid with this event order.

For rank r_x and number of uses d_x , the unmatched histogram is

$$H^{(0)}(z) = \sum_{s \text{ source}} d_s z + \sum_{x \text{ addition}} \left((d_x - 1)z^{r_x} + z^{h-r_x} + \sum_{y \text{ operand of } x} z^{r_x-r_y} \right) \\ + \sum_{x \text{ side}} (z^{h-1-r_x} + z) + \sum_{x \text{ total}} (z^{r_x} + z^h).$$

For a match $u \rightarrow w$ preserving y , add

$$-z^{h-r_u} - z^{r_y} - z^{r_w-r_y} + z^{r_w-r_u}.$$

Rank-zero terms are bookkeeping. The selected positive-frame counts are

h	v_h	c_h	q_h	$ \mathcal{M}_h $	R_h	loss
32	4960	105056	14912	3296	116672	992
30	4060	85440	12210	2910	94740	870
40	9880	214480	29680	5120	239040	1560

The compatible histograms in the accompanying finite certificate satisfy $\sum_r rH_{h,r} = hR_h + 2h(h-1)$. They count auxiliary internal transitions only. Physical-data growth and outer transitions are accounted for separately.

2.5 Complementary auxiliary gauges

A framed scalar word telescopes to $D_{\text{out}}SD_{\text{in}}^{-1}$. On an auxiliary role restored individually by S , its endpoint action is therefore $D_{\text{out}}D_{\text{in}}^{-1}$ for every initial array. Since $D_{I-P}D_P = D_I$, a nonzero source gauge needs no physical scratch preparation. All data terminal gauges keep their prescribed values.

Set $a = 32, b = 30, k = 40$, $H = ab = 960$, and $m = Hk = 38400$. In the stage-two invocation put

$$P_0 = (I - P_a) \otimes I_b \otimes P_k, \quad Q = P_a \otimes I_b \otimes P_k, \quad P_1 = P_0 + Q.$$

Here $P_0Q = QP_0 = 0$, and the ranks are $H - b, b, H$. Use source D_{P_0} and sink D_{I-P_0} on every auxiliary role. The early mixer is at P_0 . For the inverse middle shear, raise the physical source to P_0 before its initial inverse copy; this moves an existing enlargement earlier without changing rank mass. Every later producer frame contains P_0 , including fanout and carrier continuations. An inserted conceptual P_0 vertex splits nested growth additively. Thus the entrance disappears for every real auxiliary role. The cleanup is at P_1 , and its final edge is

$$D_{I-P_0}D_{P_1} = D_ID_Q = D_{I-Q}.$$

Its rank is $m - b = (H - b) + (m - H)$, preserving the boundary rank budget. The arbitrary-input scalar cancellation above also applies to the changed logical interpretation of the auxiliary arrays.

2.6 Bank and rank accounting

Put $N = v_a v_b v_k$ and $B_i = NR_{h_i}/v_{h_i}$, with $(h_1, h_2, h_3) = (a, b, k)$. The intersection-one triple graph has a regular bipartite double cover of positive degree. Edge counting gives Hall's condition and a perfect matching. Using the resulting middle-axis neighbor permutation, share all stage-one roles

with stage-three roles; their endpoint spaces are orthogonal where needed. There are B_1 paired roles of rank $m - a - k$ and $E = B_3 - B_1$ unmatched stage-three roles of rank $m - k$. Precisely,

$$\begin{aligned} N &= 198959488000, & B_1 &= 4680040601600, \\ B_2 &= 4642714752000, & B_3 &= 4813691904000, \\ E &= 133651302400, & W &= 2N + B_2 + B_3 = 9854325632000, \\ L &= N \left(\frac{6}{a-2} + \frac{6}{b-2} + \frac{6}{k-2} \right) = 113840729600. \end{aligned}$$

Consequently

$$s = Wm - 2N + 2L = 378405934031283200, \quad Wm - s = 170237516800.$$

The direct stage-two entrance applies to the full B_2 bank, and its physical-data residual occurs $2N$ times.

3 A simultaneous controlled basis

3.1 The ordinary projector corner

For a rank- j projector in dimension h with $j > h/2$, put $r = h - j$. In a generic rational basis its first- r /last- r corner is invertible: one witness pairs these coordinates in rank-one blocks $\frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$ and is the identity on the middle $h - 2r$ coordinates. Similarity of rational projectors with the same rank proves that the relevant minor is a nonzero rational function of the basis.

Write $I - P = XY$, $YX = I_r$. The corner is $-X_L Y_R$, so X_L, Y_R are invertible. Eliminating it leaves on the middle coordinates

$$I - X_M Y_M - (-X_M Y_R)(-Y_R^{-1} X_L^{-1})(-X_L Y_M) = I_{h-2r}.$$

The resulting lower-lower profile is r singleton pivots and one contiguous block of width $2j - h$. For $j \leq h/2$ retain j singleton pivots; for $j = h$ use one block of width h . The last is still smaller than the full tensor dimension m . Nonzero scalings are absorbed in the triangular factors. This argument is also the large-projector identity used in the controlled corners below.

3.2 The selected inner family

Use physical index $i = b\alpha + \beta$, with β fastest, and

$$K = T_2(I_a \otimes L_b), \quad T_2(v \otimes e_\beta) = J_\beta v \otimes e_\beta, \quad J_\beta = M_\beta L_a.$$

The bases L_a, L_b are free invertible rational matrices. For $I = (0, \dots, a-1)$ prescribe the first $2a$ row labels as $[I, (1, \dots, a-1, 0)]$ and the last $2a$ inverse-column labels as $[(a-1, 0, \dots, a-2), I]$. A row label c at (α, β) means $e_\alpha^t M_\beta = e_c^t$; an inverse-column label means $M_\beta^{-1} e_\alpha = e_c$, equivalently a prescribed column of M_β . Their effective residues $\beta - c$ modulo b are $\{0, 1, 3\}$ and $\{25, 27, 28\}$, respectively. Within one side no coefficient coordinate is repeated at fixed β ; the two residue sets and the physical boundary positions are disjoint. Each M_β thus has a permutation completion. These linear constraints define an affine space with a nonempty irreducible invertible locus.

The first a rows and last a inverse columns match the coordinate orders of L_a, L_a^{-1} , giving every ordinary axis- a corner. For axis b , the inactive first factor is rank one and its weights separate, giving a diagonal rescaling of $L_b B L_b^{-1}$. These facts apply to every internal producer projector and the data residuals.

3.3 The A5 tree witnesses

The inner projector $(I - P_a) \otimes (I - P_b)$ has nullity $a + b - 1 = 61$. After eliminating its first b nullspace coordinates and rescaling nonzero line entries, its forward restriction is the incidence matrix on vertices $0, \dots, b + 1$ with edges

$$(0, b), (1, b + 1); \quad (j + 2, j + 1) \quad (0 \leq j \leq b - 3); \quad (0, b - 1).$$

It is the path on $1, \dots, b - 1, 0$ with $b, b + 1$ attached at the ends. The dual restriction has edges

$$(b, 0), (b + 1, 1), (b + 1, 2); \quad (j - 2, j - 1) \quad (4 \leq j \leq b + 1),$$

the path $0, b, b - 1, \dots, 2, b + 1, 1$. Each tree has $a - 1$ edges; leaf elimination gives rank $a - 1$. Both nullspace restrictions are therefore nonsingular. The large-projector identity gives 61 singleton pivots and one 838-block.

3.4 Outer matching and the A1 affine witness

Set

$$S = T(I_k \otimes K), \quad T(u \otimes e_i) = G_i u \otimes e_i.$$

For $i = 0, \dots, k - 1$ prescribe the first row of G_i as row i of a free L_k . At the last k multiplicity indices prescribe the last inverse columns in matching L_k^{-1} order. The next a first rows and preceding a inverse columns remain free. The boundary intervals of length $a + k = 72$ are disjoint. Ordinary axis- k corners and the full k -block for unmatched stage-three roles are retained.

The paired A1 null projector has the form

$$Q_1 = I_k \otimes P_q + P_t \otimes P_U, \quad \text{rank } P_q = 1, \quad \text{rank } P_U = a, \quad P_q P_U = 0.$$

Write q_i for the transformed line coordinate and U_i for the transformed U row, impose $q_i \neq 0$, and put $z_i = U_i/q_i$. Absorbing L_k into outer coordinates, the first k nullspace rows are $[e_p^t, t_p z_p]$, with $t_p \neq 0$. The first a vectors z_p are nonzero multiples of the L_a coordinate basis. Row $p = a$ repeats coefficient 1 at $\beta = 2$ instead of $\beta = 1$; its multiplier is generically different. Indeed the two distinct motif lines have coordinate ratios not identically equal in these freely chosen L_b coordinates. Thus $z_0, \dots, z_a$ affinely span $\mathbb{Q}^a$. For each extra row with free first-row functional g_i , eliminating the first k coordinates leaves

$$\sum_{p < k} g_{i,p} t_p (z_i - z_p).$$

Affine spanning makes this map onto $\mathbb{Q}^a$. Choose the a extra rows to produce its coordinate basis. This witnesses a nonzero first-72-row nullspace determinant. Dually the last a columns give the matching L_a^{-1} basis and the preceding column repeats coefficient $a - 2$ at $\beta = b - 3$ instead of $b - 2$. The same affine argument and the free additional inverse columns witness the dual determinant. The disjoint outer intervals make these prescriptions compatible. Hence A1 has 72 singleton pivots and a 38256-block.

3.5 The gauged and A3 corners

For the stage-two null projector $Q = P_k \otimes (P_a \otimes I_b)$, the first- b /last- b corner of SQS^{-1} is diagonal. With $i = (0, \beta)$ and $j = (a - 1, \beta)$ its diagonal entry is

$$(e_1^t G_i t_k)(\phi_k G_j^{-1} e_k)(e_1^t J_\beta t_a)(\phi_a J_\beta^{-1} e_a),$$

where $P_h = t_h \phi_h$ and $\phi_h(t_h) = 1$. Each factor is generically nonzero in the prescribed family. The corner of $I - Q$ is its negative; it gives one contiguous b -block, and the Schur identity leaves $m - 2b$. Thus its profile is (30, 38340).

The A3 null projector is $P_t \otimes I_H + (I - P_t) \otimes P_q$, of rank $H + k - 1$. Its first H rows restrict diagonally and nonsingularly to the first summand. After eliminating them, each of the next $k - 1$ rows is a nonzero q_i multiple of

$$\text{row}_2(G_i) - \frac{\text{row}_2(G_i)t}{\text{row}_1(G_i)t} \text{row}_1(G_i)$$

on the complement of t . These second rows are free and can give any basis of that complement; the dual witness uses free next-to-last inverse columns. Both full nullspace restrictions are nonsingular. Within the H -corner, the matched first- k /last- k corner is a diagonal rescaling of an ordinary rank- $(k-1)$ projector. Its one singleton and $(k-2)$ -block exhaust that perturbation, leaving the matching interval of width $H - 2k + 2$. The remaining corner uses $2(k-1)$ singleton pivots. The final profile is therefore 79 singletons and blocks 38, 882, 36402.

All required determinants, ordinary corner minors, nonzero coordinates, and unequal ratios are nonzero rational functions on the same irreducible family. The displayed witnesses prove each is nonzero; after clearing denominators, their finite product is nonzero. Rational enumeration therefore finds one basis satisfying them all, independently of recursive width. Choose a sufficiently large odd address prime avoiding its finitely many bad minors and denominators.

3.6 Complete bit characteristic

The complete macro and physical-data profiles are

class	copies	child widths
paired A1	B_1	72 singletons; 38256
unmatched stage three	E	40; 38320
gauged stage two	B_2	30; 38340
A3 data	$2N$	79 singletons; 38; 882; 36402
A5 data	$2N$	61 singletons; 838
stage-one data	$2N$	1; 30
stage-two data	$2N$	1; 28
stage-three data	$2N$	1; 38

For each axis- h rank- r histogram entry use its ordinary profile and multiplicity $NH_{h,r}/v_h$. This gives 629339590854400 singleton calls, with $\sum_t n_t t = s$ and every $0 < t < m$. Set

$$\Psi_b(x) = \frac{\sum_t n_t t^x}{Wm^x}, \quad a_b = \frac{2993093}{250000000000}, \quad \tau = 1 - a_b.$$

For $1 \leq y \leq 2$ and $z = (y-1)/(y+1)$ the positive atanh series gives

$$\log y \leq 2 \sum_{j=0}^{23} \frac{z^{2j+1}}{2j+1} + \frac{2z^{49}}{49(1-z^2)}.$$

Extract powers of two and round this bound upward to 10^{-12} , obtaining $\ell_t \geq \log(m/t)$. For $0 \leq u < 3$, $e^u \leq 1 + u + u^2/[2(1-u/3)]$. Exact rational arithmetic then gives

$$\Psi_b(\tau) \leq \sum_t \frac{n_t t}{Wm} \left(1 + a_b \ell_t + \frac{(a_b \ell_t)^2}{2(1 - a_b \ell_t/3)} \right) < 1 - \frac{226731361147908}{10^{28}}.$$

The strict characteristic and the integer-width proof below establish the bit interface with exponent τ .

4 Complete rows and integer widths

The following mixed-width proof applies with the child list just established; the ordered-affine wrappers are the triangular conjugations above.

4.1 Integer widths, bounded row padding, and mixed contractions

The batched residual construction uses a fixed finite list of child sizes, not only powers of one radix. The following implementation supplies the corresponding integer-width recursion. All constants in this subsection are fixed finite circuit constants, independent of the input length.

A general moment condition. Let $m, W \geq 2$ be fixed. At one circuit invocation, let the list $t_1, \dots, t_J$ contain integers in $[1, m-1]$. A child of type i has width $t_i f$, where $f = \lfloor e/m \rfloor$, and operates on exactly $1/W$ of the parent's logical volume. Repetitions in the list represent separate calls. An ordinary rank-one call has $t_i = 1$; a batched block has the corresponding larger value of t_i . Full-rank edges may retain their original m rank-one calls, so no same-width recursive call is needed. Put

$$\Psi(x) = \frac{1}{W} \sum_{i=1}^J (t_i/m)^x, \quad t_* = \max_i t_i < m.$$

If $\Psi(\tau) < 1$ for some $0 < \tau < 1$, then the normalized recurrence

$$F(e) \leq \frac{1}{W} \sum_{i=1}^J F(t_i \lfloor e/m \rfloor) + C, \quad F(e) = O(1) \quad (1 \leq e < m),$$

has $F(e) = O(e^\tau)$ for every positive integer e . Indeed, substituting $F(u) \leq Au^\tau$ into an internal call gives at most $A\Psi(\tau)e^\tau + C$. Choose A to cover the finite base range and $C/(1 - \Psi(\tau))$. Floors can only decrease the right-hand side.

A finite integer depth bound. Define

$$D(e) = 0 \quad (e < m), \quad D(e) = 1 + D(t_* \lfloor e/m \rfloor) \quad (e \geq m).$$

This is an integer algorithm: repeatedly replace e by $t_* \lfloor e/m \rfloor$ until it is below m . The function D is nondecreasing, every child has depth bound at most $D(e) - 1$, and $D(e) = O(\log(2e))$, because each internal replacement is at most $(t_*/m)e$. A row count divisible by $W^{D(e)}$ is therefore sufficient for every root-to-leaf path. At each invocation write the row index as $Wg + w$ and split by w . Every child has complete within-row ranges, logical volume exactly V/W , and a remaining row count divisible by $W^{D(e)-1}$. No padding occurs inside this recursion.

Bit interchange with a remainder. At an integer-width call write $e = mf + r$, $0 \leq r < m$. If $f = 0$, interchange the bounded number of digits individually. Otherwise separate the high r digits of each of the two target chunks. With G denoting all intervening spectators, the layout is

$$(H^+, H^-, G, D^+, D^-).$$

First exchange corresponding digits of H^+, D^+ and move H^+ just before H^- . The layout becomes (D^+, H^+, H^-, G, D^-) . Apply the finite circuit to the two length- mf chunks H^-, D^- . Finally move H^+ back immediately before the final H^- , obtaining

$$(D^+, D^-, G, H^+, H^-).$$

All other fields are preserved. These moves use fewer than a fixed multiple of m single-digit operations and cost $O(V)$. The main chunks split into m equal width- f fields, so every rational shear and every batched contiguous-block shear has exactly the required field shape. A child retains the unused main digits and all remainders as spectators. Thus the preceding moment recurrence applies literally at every integer width; there is no per-level numerical padding of either chunk range.

Supplying the bit rows from existing digits. For an arbitrary width e , compute $D(e)$ and choose the least nonnegative integer ρ such that

$$q^{2\rho} \geq W^{D(e)}.$$

Both numbers can be constructed by fixed-radix integer arithmetic. Since $D(e) = O(\log e)$, we have $\rho = O(\log e)$. For sufficiently large e , $2\rho < e$. The exceptional set of smaller widths is fixed and finite; individual-digit interchange there is absorbed by the asymptotic constant.

Exchange the high ρ digits of the two chunks, then gather their two high parts into one row field before both remaining chunks, exactly as in the preceding layout operation. This row field has range $q^{2\rho}$. Within each original prefix, pad its range to

$$R' = W^{D(e)} \left\lceil q^{2\rho} / W^{D(e)} \right\rceil < 2q^{2\rho}.$$

Every added row includes the entire remaining address set. Recurse once at the remaining width $e - \rho$, which has depth bound at most $D(e)$; then remove the added rows and undo the gathering move. Each completed invocation preserves its row index, so padded zero rows are zero again when they are removed.

The possibly large exponent in $W^{D(e)} = e^{O(1)}$ does not multiply the logical volume by that polynomial. The row digits were already part of the two input chunks. Only rounding the row count adds data, by a factor strictly below two, once at the outer boundary. Exchanging and gathering $O(\log e)$ high digits costs $O(V \log(2e))$, which is $O(Ve^\tau)$. The radix conversion from binary chunk ranges to the fixed radix q retains the original constant factor, less than q^2 . Consequently the batched construction is an arbitrary-width interchange procedure with the asserted exponent τ and no additional power of its volume or width.

Fixed tapes and descriptors. Use the original depth-first scheduler: park only the other role streams at the active stack top, run the child on one complete role stream, and restore the parked streams on return. The number of child types is fixed, so parking, copying, cleanup and the finite arithmetic transforms contribute $O(V)$ per node. Before descending, retain the two current chunks and merge consecutive spectator fields; the descriptor has a fixed number of field intervals. Its integer lengths have $O(\log V_0)$ bits. All descendants retain the complete ranges of both root main chunks, even when most digits have become spectators. Their volume is therefore at least $q^{2(e-\rho)}$ in the outer construction, while $j \log W = O(\log e)$ at depth j . Hence $\log V_0 = \log V_j + j \log W = O(\log(2V_j))$. Polynomial descriptor work is absorbed by $O(V_j)$, as in the original fixed-tape proof. The larger fixed constant in the depth bound creates no new tape and no growing alphabet.

5 Complex residuals and precision

5.1 A normal form for every nondegenerate binary residual

Let E have binary basis matrix V and nonsingular Gram matrix $G = V^t V$. In dual Fourier coordinates put $q(z) = \text{wt}(VG^{-1}z) \pmod{4}$. Its polarization is $2z^t G^{-1}w$. The same argument applies to mixed signs on orthogonal summands, since reversing a sign preserves this polarization. Write $S_q = \sum_z i^{q(z)}$. Completing the square gives

$$\begin{aligned} K(x, y) &= 2^{-r} \sum_z i^{q(z)} (-1)^{z^t(x+y)} \\ &= 2^{-r} S_q i^{-q(Gx)} (-1)^{x^t G y} i^{-q(Gy)}. \end{aligned}$$

The scaled Walsh kernel in the middle is a binary permutation of $H_0^{\otimes r}$, where

$$H_0 = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = b S C S, \quad b = (1 - i)/2, \quad S = \text{diag}(1, i).$$

The remaining scalar $S_q b^r$ is a fourth root of unity. To verify this, decompose the nondegenerate quadratic module into odd one-dimensional and alternating two-dimensional blocks. Their Gauss sums are $1 \pm i$ and ± 2 , respectively; multiplying by b and b^2 gives a fourth root in each case. This is a proof device, not an orthonormal-basis assumption on E .

Consequently one rank- r child suffices, with binary address changes, diagonal fourth-root phases, and one scalar fourth-root phase. On f parallel columns it is one $C^{\otimes rf}$ child, the scalar raised to power f , and columnwise quadratic phases modulo four. The retained packed address operations evaluate these fixed formulas. These wrappers have modulus one and introduce no denominator loss.

5.2 The selected Gaussian-dyadic producer

Use paired triple exclusion with base threshold two, complete unsplit disjoint and pair-star sums, and sharing by actual ordinary support. For an ordinary support with at least two common points use the span of its triple indicators; these form an orthonormal common-pair star. Otherwise use the coordinate space of its union. Support inclusion preserves these labels: losing the common pair enters a containing coordinate space. Source lines use ordinary triple containment.

Retain every full pair-star $P_{uv} = \sum_{w \neq u, v} X_{uvw}$ and form $B_u = \sum_{v \neq u} P_{uv} = 2 \sum_{T \ni u} X_T$. For even h label these central additions by the nondegenerate hyperplane $E_u = \{x : \sum_{j \neq u} x_j = 0\}$; every contributing star lies in it. Retain the global total T with full-space label. The central correction at target S is

$$\frac{1}{4} \sum_{u \in S} B_u - \frac{1}{2} T = \sum_U \frac{|S \cap U| - 1}{2} X_U.$$

Adding half the disjoint sum and subtracting half the intersection-two sum gives exactly X_S . All coefficients are Gaussian dyadic. The transparent retained-output wrapper and oriented carrier compiler apply to these nested nondegenerate labels. The h hyperplane centers lose $h(h-1)$ and the global center loses h , so loss is h^2 . The selected counts, including all point centers, are

h	v_h	c_h	q_h	$ \mathcal{M}_h $	R_h	loss
32	4960	122512	19873	19241	123144	1024
30	4060	99153	16271	15873	99551	900
40	9880	252023	39561	37642	253942	1600

Their internal histograms satisfy $\sum_r r H_{h,r} = h R_h + 2h^2$. They are distinct from the bit histograms.

5.3 Endpoint normalization and the complex characteristic

For nondegenerate U , Fourier phases give

$$C_U^2 = X_{P_U \mathbf{1}}, \quad C_{U^\perp} C_U^{-1} = C_F X_{P_U \mathbf{1}}.$$

Put $v = P_U \mathbf{1}$. On coordinates in v , this endpoint has C^{-1} instead of C . The identity $C^{-1} = -iZCZ$ shows that conjugating by Z_v and multiplying by $i^{\text{wt}(v)}$ normalizes it to C_F ; for f columns the exponent is $f \text{wt}(v)$. These are diagonal wrappers without a child.

Choose stage-two source C_{D_0} and sink $C_{D_0^\perp}$, where $D_1 = D_0 \perp A$ and $\dim A = b$. The same refined copy schedule removes the entrance on every auxiliary role. With $R = D_1^\perp$, the exit is $C_R C_{D_0}^{-1}$ on $E = R \perp D_0 = A^\perp$. Its rank is $m - b$ and its nondegenerate polar form is unchanged by the mixed signs. The normal form above executes it as one child. Arbitrary auxiliary restoration justifies the endpoint normalization on the entire B_2 bank.

Use the complex counts to define $B_i = NR_{h_i}/v_{h_i}$, and now

$$\begin{aligned} W &= 2N + B_2 + B_3 = 10390178240000, \\ L &= \sum_i (N/v_{h_i}) h_i^2 = 117399987200, \\ s &= Wm - 2N + 2L = 398982681296998400. \end{aligned}$$

The whole-residual macro list is

t	38328	38360	38370	37401	899
n_t	B_1	$B_3 - B_1$	B_2	$2N$	$2N$.

Add $(N/v_h)H_{h,r}$ rank- r children for each positive internal rank. Leave the remaining physical-data rank, 39393978624000, as singleton calls. The resulting list has weighted rank exactly s . For $a_c = 3/125000$ and $\sigma = 1 - a_c$, the rational logarithm bounds above and $e^u \leq (1 - u)^{-1}$, $0 \leq u < 1$, give

$$\Psi_c(\sigma) \leq \sum_t \frac{n_t t}{Wm(1 - a_c \ell_t)} < 1 - \frac{792981575160764}{10^{23}}.$$

5.4 The dependency-path guard

Before gauging, nested producer labels telescope along every arithmetic path, including wire switches at common-frame scalar gates. Retained center drops occur on one return frontier per stage, crossed at most once by a path, and each costs at most that axis dimension. Carrier reuse adds only increasing continuations. This gives $q_0 = m + 2(a + b + k)$. Gauging removes an early edge and adds $ab - b$ at a late stage-two edge. A path meets at most one such edge, since it terminates that stage-two role, which is not shared across stages. Thus the conservative total rank budget is

$$q = m + 2(a + b + k) + ab = 39564.$$

Every child except the four large macros is at most 899; two large macros cannot fit within q . Set $\rho = 19/10$, $\gamma = 9997/10000$. The integer comparisons $m^{19} > 5000000000^{10}$ and $899^9 < 460^{10}$ give $m^\rho > 5000000000$ and $899^{\rho-1} < 460$. If a path has one large child t , its moment is at most

$$(t/m)^\rho + (q - t)899^{\rho-1}/m^\rho.$$

This expression increases over the large-child range, since its numerator derivative is $\rho t^{\rho-1} - 899\rho^{-1} > 0$. Use $t = m - b$ and the concavity bound $(1 - x)^{19/10} \leq 1 - 19x/10 + 9x^2/10$ to obtain

$$\sum_{\text{path}} (r/m)^\rho < 1 - \frac{19}{12800} + \frac{9}{10 \cdot 1280^2} + \frac{1194 \cdot 460}{500000000} = \frac{51180270301}{51200000000} < \gamma.$$

Without a large macro the bound is $227493/6250000 < \gamma$.

For coefficient-depth accounting take $E_0 = 128(W + m + 1)^3$. There are at most s positive-rank residual calls and W endpoint normalizations. Combining the quadratic phases with the S factors, each uses at most two diagonal phase passes and one scalar phase. The inherited gate and remainder count is therefore bounded conservatively by

$$36W^3 + 16s + 16W + 8m + 4 < E_0.$$

Binary address changes only permute coefficients; descriptor evaluation is paid by the retained packed record work and does not increase coefficient depth. Stop below width d^β , where $\beta = 1/1000$. For $e = mf + r$, $0 \leq r < m$, recurse on tf and treat the remainder individually. For sufficiently large d , a leaf descending from an internal node has width at least $d^\beta/(2m)$. Choose $C_{\text{dep}} \geq 16m$ and $(1 - \gamma)C_{\text{dep}} \geq E_0 + 16m + 1$. The retained elementary-kernel bound $A(e) \leq 8e$ and strong induction using the path moment imply

$$A(e) \leq C_{\text{dep}} d^{-\beta(\rho-1)} e^\rho.$$

At a leaf the lower width bound proves this inequality, while at an internal node the unused $1 - \gamma$ fraction covers local work. Small roots are handled directly. With $\zeta = 1/10000$ take

$$C_1 = \rho - (\rho - 1)\beta + \zeta = \frac{1187}{625}, \quad C_0 = \lceil 128m(1 + 1/\zeta)C_{\text{dep}} \rceil, \quad \Delta = \lceil C_0 d^{C_1} \rceil.$$

The retained exact denominator/magnitude-depth induction gives $p + \Delta$ fractional bits and $\Delta + 3$ integer/sign bits. All selected scalar coefficients are Gaussian dyadic and the new wrappers have modulus one, so that induction still applies. For $\epsilon C_1 < 1$ their total width is $O(p)$.

5.5 Rows, stopping, and transfer to the layer interface

Let $t_* < m$ be the largest child and compute $D(e) = 0$ for $e < m$ and $D(e) = 1 + D(t_* \lfloor e/m \rfloor)$ otherwise. It is $O(\log e)$ and bounds every descendant depth. With the retained compact-control parameters, reserve

$$q_0 = \lceil \log_2 W \rceil D(d), \quad Q_0 = q_0 + q_F + q_B.$$

Process the first $q_0 + q_F$ and last q_B selected axes individually; if at most Q_0 axes exist, process them all individually. Otherwise the middle active interval is consecutive. Only the q_0 row chunks supply row indices, separate from the two compact fields. Pad their count once to a multiple of $W^{D(d)}$, by a factor below two. Every child retains complete compact and spectator ranges and has exactly $1/W$ of its parent's volume. No internal padding is needed. The reserved-axis bound remains

$$Q_0 = O(\log(2d) + d^{\max\{1-c, 0\}} \log p + 1).$$

Remainders commute with the completed tensor map and cost $O(V)$ per node. The fixed-tape depth-first scheduler and complete-row argument of the bit construction apply with the retained compact descriptors.

Since $\Psi_c(\sigma) < 1$, the total leaf σ -weighted width, including the volume factor W^{-j} at depth j , is at most d^σ . Thus leaf work is $O(d^{\sigma+\beta(1-\sigma)})$. Internal compact operations cost $O(G^\tau e^\tau + 1)$ per current volume. Summing the strict moments gives the retained complete normalized layer bound

$$O\left((\log p)^3 \left[d^\chi + d^{\sigma+\beta(1-\sigma)} + d^{\max\{1-c,0\}} + 1\right]\right), \quad \chi = \tau + (1 - \beta) \max\{\sigma - \tau, 0\}.$$

For $\sigma \leq \tau$, use $\Psi_c(\tau) \leq \Psi_c(\sigma)$; otherwise use $e^\tau \leq d^{\beta(\tau-\sigma)} e^\sigma$ at each internal node. These arguments do not assume equal leaf depth. The usual strict conditions on λ, λ' therefore give the retained $O(Vd^{\lambda'})$ layer interface, now with the new moment and the new C_{dep} .

6 Conditional assembly

Theorem 6.1 (Conditional fixed-tape multiplication bound). *Assume the pinned upstream multiplication theorem and retained analytic, ordered-affine, compact-control, Gaussian-map, and fixed-tape interfaces specified in the opening section. The endpoint-gauged interchange interface and the complex layer proved here may be substituted into those interfaces. They give*

$$T(n) = O\left(n(\log n)^{1-\kappa}\right), \quad \boxed{\kappa = \frac{119723}{20000000000} = 5.98615 \times 10^{-6}}.$$

Proof. Take the following exact parameters:

$$\begin{aligned} a_b &= \frac{2993093}{250000000000}, & a_c &= \frac{3}{125000}, \\ \tau &= 1 - a_b, & \sigma &= 1 - a_c, \\ \epsilon &= \frac{1}{2 + a_b} = \frac{250000000000}{500002993093}, & c &= 1, \\ \beta &= \frac{1}{1000}, & \delta &= \frac{1}{10^{16}}, \\ \lambda &= \tau + \frac{1}{10^{20}}, & \lambda' &= \tau + \frac{2}{10^{20}}, \\ \zeta &= \frac{1}{10000}, & C_1 &= \frac{1187}{625}. \end{aligned}$$

The strict bit and complex moments were certified above. Since $\sigma < \tau$, the internal layer exponent is $\chi = \tau$. The complex leaf saving is

$$(1 - \beta)a_c = \frac{2997}{125000000} = 2.3976 \times 10^{-5} > a_b.$$

Thus both the internal and leaf exponents are strictly below λ' , and the displayed gaps between τ, λ, λ' absorb fixed logarithmic factors. The precision guard has $C_1 = 19/10 - 9\beta/10 + \zeta$ and $1 - \epsilon C_1 > 0$. Compact control allows $c = 1$; its reservation exponent is zero and $1 - \epsilon(1 + c) = 1 - 2\epsilon > 0$ gives $K = o(\ell)$ with $K = d$.

For the retained Gaussian parameters $\alpha = \lfloor \sqrt{b/(8d)} \rfloor$ and $\gamma_G = 2d\alpha^2 \leq b/4$, the same positive $1 - 2\epsilon$ ensures the eventual conditions $b \geq 64d^2$, $\alpha^2\theta_i \geq 1$, and prime-interval growth. Use the corrected chirp input bound $F = 2^{\lceil 1.14\alpha^2 \rceil}$; the following section proves that $P = 34p$ still suffices. The retained target errors and final rounding therefore apply.

The seven assembly savings are the exact rational numbers

$$\begin{aligned} g_1 &= 1 - 2\epsilon, & g_2 &= \epsilon a_b, & g_3 &= \epsilon(a_b - 2/10^{20}), \\ g_4 &= (1 - \epsilon)a_b, & g_5 &= 1 - \delta - 2\epsilon, & g_6 &= 1 - \delta - \epsilon, & g_7 &= \epsilon. \end{aligned}$$

Writing $b_0 = a_b/(2 + a_b)$ gives $g_1 = g_2 = b_0$, $g_3 = b_0 - 2\epsilon/10^{20}$, $g_4 > b_0$, and $g_5 = b_0 - \delta$. Since $\delta > 2\epsilon/10^{20}$, g_5 is the least of the seven. Exact rational subtraction gives

$$\min_i g_i - \kappa = \frac{a_b}{2 + a_b} - \frac{1}{10^{16}} - \frac{119723}{20000000000} = \frac{828463377506907}{5000029930930000000000000000} > 0.$$

All retained parameter conditions are strict with this choice; in particular the finite arithmetic certificate evaluates the complete 29-condition interface and these seven margins with rational arithmetic. The final positive gap absorbs remaining logarithmic factors, proving the claimed bound in the retained model. $\square$

The count and moment data are finite arithmetic certificates for the specified producer and common-basis construction. The theorem's use of the pinned analytic interfaces and its fixed rational basis preparation is separate from checking those integers and rational inequalities.

7 Gaussian input-error correction

The retained fast Gaussian construction generates $2^{-B}e^{\pi\alpha^2\beta_j^2}$ at precision P , where $B = \lceil 1.14\alpha^2 \rceil$, then shifts by B and multiplies by the disk-grid input u_j . The exponential interface gives absolute error at most 2^{1-P} before the shift. Thus the error after shifting is controlled by 2^{B+1-P} , not by a constant multiple of $2^{-P}e^{\pi\alpha^2/4}$ uniformly in α .

Set $F = 2^B$ in the retained chirped-correlation block lemma. Since $|\beta_j| \leq 1/2$ and $\pi/(4\log 2) < 1.14$, this bounds the exact input magnitude. Including the product and rounding, it also gives the required input accuracy $|\tilde{f}_j - f_j| \leq 2^{4-P}F$. The distinction matters: $2^B/e^{\pi\alpha^2/4}$ is unbounded, so the smaller exact-magnitude bound alone does not justify this accuracy.

No additional asymptotic precision is needed. For the inherited integer parameters $p > 100$ and $2 \leq \alpha < \sqrt{p}$, let $L_A = 3\lceil \sqrt{p}/(2\alpha) \rceil + 1$ and $\Gamma = 29p$. Then $L_A \leq p$, $\lceil \log_2 L_A \rceil \leq p - 1$ and $B \leq 1.14p + 1$. The block lemma requires

$$P \geq p + \Gamma + \lceil \log_2(FL_A) \rceil + 11.$$

Its right side is at most $32.14p + 11 < 34p$, so the retained $P = 34p$ satisfies the corrected error budget. The packing widths remain $O(p)$, and the target errors, tape-time exponents and final rounding guarantee are unchanged. The combined manuscript patch includes this correction directly in the retained Gaussian proof.

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